

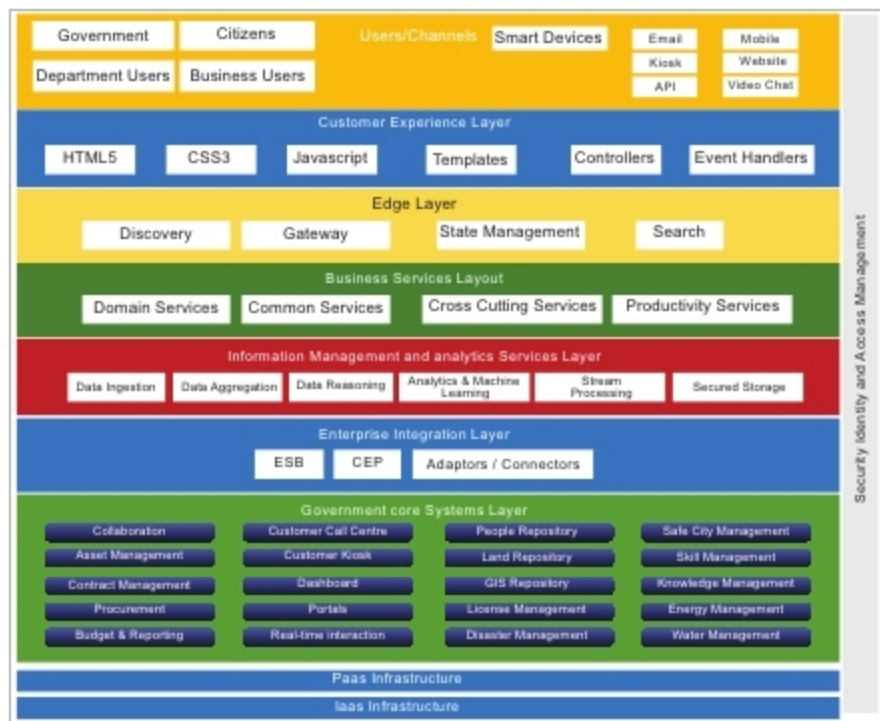


Figure 3: Overview of the digital architecture for e-governance

any duplication. Figure 3 depicts the high-level architecture for e-governance.

The adoption of new technology trends like cloud computing, IoT and mobility should also be considered while developing open source based government applications.

Open source software governments can benefit from

The following sections briefly describe the new technologies governments can adopt, and the open source software and products they can use to do that.

The cloud: Cloud based open source technology enables governments to accelerate innovation, creating and deploying applications with speed and consistency. For example, OpenStack is a set of tools that allows users to create, automate and manage both public and private clouds at minimal expense. With OpenStack, governments can set up their own cloud to hold data they do not want in a public cloud.

Application platforms handle complex connectivity issues behind the scenes (including automatic documentation), letting users concentrate on creating the right workflows and business logic, but also making it easier and more cost-effective for organisations to manage frequent change.



Figure 4: OSS adoption for e-governance

It is imperative to make sure that the integration solution is optimised to connect the new applications in the cloud environments with older applications, systems, services and databases, no matter where they are located.

Open application programming interfaces (APIs):

The application architecture should use open APIs to enable quick and transparent integration with other e-governance applications and systems implemented by various government organisations, thereby providing access to data and services, as well as promoting citizen/developer participation for the benefit of the community.

API management includes the full API management life cycle, i.e., integration with diverse applications and appliances to support the management, monitoring and tracking of all your API activities. It also helps to do the following:

- Enable mobile strategies
- Develop a customer or partner ecosystem
- Multiply the reach of the content, data and technology
- Create new business models
- Foster internal innovation

Large complex systems in e-governance follow an API-centric approach, exposing services for integration with multiple third party solutions. The APIs are RESTful, XML based, and stateless services. All APIs are accessible via the HTTPS protocol.

Service discoverability: This is the process of the client application querying the central registry to learn the location of services. Service discoverability ensures that metadata attached to a service describes the overall purpose of the service and its functionality, which makes the services easily discoverable. A repository of re-usable business logic components needs to be maintained and made available as part of service discoverability.

Consul, Apache Zookeeper, Etcd, Netflix Eureka, etc, are the widely used open source service discovery tools.

Platform and database agnostic: Applications should be forward and backward compatible and be deployable on any technology platform. They should be able to communicate with any data store.

For example, Jenkins is a continuous delivery tool that builds and tests software after every change. Jenkins can deploy new code to production, which helps in no down time in making the upgraded product available to constituents.

PostgreSQL, MySQL, MongoDB and CouchBase are the widely used open source databases.

Microservices: The design of the business and information services layer of the e-governance application is developed using the microservices architecture principles and cross-channel capabilities. Microservices architecture (MSA) allows the creation of services that are loosely coupled and have a different programming language base. They are scalable, have a quicker delivery time, etc. New e-governance applications need to be developed as reusable microservices based on open standards and which expose the standard recommended integration interfaces.